



3.1 Volume and surface area of 3D solids

Geometry and trigonometry · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Standard formulas: cuboid/prism $V = \text{base area} \times \text{height}$; cylinder $V = \pi r^2 h$, curved surface area $= 2\pi r h$; cone $V = \frac{1}{3}\pi r^2 h$, curved surface area $= \pi r l$ (l = slant height); sphere $V = \frac{4}{3}\pi r^3$, surface area $= 4\pi r^2$; pyramid $V = \frac{1}{3} \times \text{base area} \times \text{height}$. Total surface area sums all faces including base(s).

Worked example: follow the reasoning

Find the volume and total surface area of a cone with radius 5 cm and height 12 cm.

Slant height $l = \sqrt{5^2 + 12^2} = \sqrt{169} = 13$ cm.

Volume $= \frac{1}{3}\pi(5^2)(12) = \frac{1}{3}\pi(300) = 100\pi \approx 314.2$ cm³.

Total surface area $= \pi r l + \pi r^2 = \pi(5)(13) + \pi(25) = 65\pi + 25\pi = 90\pi \approx 282.7$ cm².

Quick check · multiple choice

A sphere has radius 3 cm. What is its volume?

A 12π cm³

B 36π cm³

C 27π cm³

D 9π cm³

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A solid consists of a cylinder of radius 4 cm and height 10 cm, topped with a hemisphere of the same radius. [6]
- (a) Find the total volume of the solid, in terms of π .
- (b) Find the total surface area of the solid (curved cylinder surface + hemisphere curved surface + one circular base), in terms of π .

Final answer: _____



Answers and explanation

Question 1 [6]

(a) Cylinder volume = $\pi(4^2)(10) = 160\pi$. Hemisphere volume = $(1/2)(4/3)\pi(4^3) = (2/3)\pi(64) = 128\pi/3$. Total = $160\pi + 128\pi/3 = (480\pi + 128\pi)/3 = 608\pi/3 \text{ cm}^3$.

(b) Cylinder curved surface = $2\pi(4)(10) = 80\pi$. Hemisphere curved surface = $2\pi(4^2) = 32\pi$. Base circle = $\pi(4^2) = 16\pi$. Total = $80\pi + 32\pi + 16\pi = 128\pi \text{ cm}^2$.

Quick check answer

B. $V = (4/3)\pi(3^3) = (4/3)\pi(27) = 36\pi \text{ cm}^3$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.